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COUNTING ELEMENTS

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Total Images + Links on a Page

Ever wondered how many **links** or **images** are on a webpage? 🤔

Playwright makes it super easy to count and iterate through ALL elements of a specific type on a page! This is useful for validation, broken link checking, and testing!

! IMPORTANT!

We'll use two powerful methods: `allInnerTexts()` for links and `all()` for images!

Task 1: Count Total Links 🔗

Let's find ALL links on Amazon homepage and count them!

What is a Link?

A link in HTML is an `<a>` tag with an `href` attribute:

```
<a href="https://example.com">Click Here</a>
```

```
<a href="/products">Products</a>
```

```
<a href="#section">Jump to Section</a>
```

Count All Links

```
await page.goto('https://www.amazon.co.in');

// Get all links with href attribute
let allLinks: string[] = await page.locator('a[href]').allInnerTexts();

// Count the links
console.log('Total number of links: ' + allLinks.length);
// Output: Total number of links: 245
```

How it works:

1. `page.locator('a[href]')` → finds ALL anchor tags with href
2. `allInnerTexts()` → gets text from all matched elements
3. `.length` → counts how many links found!

Understanding `allInnerTexts()`

`allInnerTexts()` method:

- Returns an array of strings
- Gets text content from ALL matched elements
- Perfect for counting elements!
- Return type: `string[]`



Example Output

```
// allLinks is an array like this:  
[  
  'Home',  
  'Products',  
  'About Us',  
  'Contact',  
  'Privacy Policy',  
]
```

```
// ... and so on  
]  
  
// Use .length to count  
console.log(allLinks.length); // 245
```

Task 2: Count Total Images

Now let's find ALL images on the page and count them!

 *What is an Image?*

An image in HTML is an `<img>` tag, usually with `alt` and `src` attributes:

```
  
  

```

 *Count All Images*

```
// Get all images with alt attribute  
let allImages: Locator[] = await page.locator('//img[@alt]').all();
```

```
// Count the images
console.log('Total number of images: ' + allImages.length);
// Output: Total number of images: 87
```

🎯 *How it works:*

1. `//img[@alt]` → XPath to find ALL img tags with alt attribute
2. `all()` → returns array of Locator objects
3. `.length` → counts how many images found!

Understanding all() Method

`all()` method:

- Returns an array of Locators
- Each Locator represents one element
- Perfect for iteration and counting!
- Return type: `Locator[]`

💡 *Key Difference:*

`allInnerTexts()` → returns *string*[] (just text)

`all()` → returns *Locator*[] (actual element references)

Iterating Through Images

Once we have the array of images, we can loop through them and get their details!



Loop Through Images

```
// Get all images
let allImages: Locator[] = await page.locator('//img[@alt]').all();

// Loop through each image and print alt text
for (let e of allImages) {
  console.log(await e.getAttribute('alt'));
}

// Output:
// "Company Logo"
// "Product Image 1"
// "Product Image 2"
// "Sale Banner"
// ... and so on
```



What's happening:

1. We get array of all image Locators
2. We use `for...of` loop to iterate
3. For each image, we get its 'alt' attribute
4. We print the alt text to console

Complete Example

Amazon Homepage Analysis:

```
await page.goto('https://www.amazon.co.in');

// Count all links
let allLinks: string[] = await page.locator('a[href]').allInnerTexts();
console.log('Total number of links: ' + allLinks.length);
// Output: Total number of links: 245

// Count all images
let allImages: Locator[] = await page.locator('//img[@alt]').all();
console.log('Total number of images: ' + allImages.length);
// Output: Total number of images: 87

// Print all image alt texts
console.log('\nImage Alt Texts:');
for (let e of allImages) {
    let altText = await e.getAttribute('alt');
```

```
console.log('- ' + altText);
}

// Sample Output:
// - Amazon Logo
// - Prime Video
// - Electronics Sale
// - Fashion Banner
// - ...
```

CSS vs XPath Selectors

Element	CSS Selector	XPath
All Links	a[href]	//a[@href]
All Images	img[alt]	//img[@alt]
All Buttons	button	//button
All Inputs	input	//input



REMEMBER!

→ `allInnerTexts()` returns `string[]` - good for links

- `all()` returns `Locator[]` - good for images/elements
- Use `.length` to count elements
- Use `for...of` loop to iterate through arrays

Real-World Use Cases

1 Broken Link Detection

Get all links → Check if each link is valid → Report broken ones

2 SEO Validation

Check all images have alt attributes → Important for accessibility!

3 Performance Testing

Count total images → Check if page has too many images

4 Content Validation

Verify expected number of links/images on critical pages

X COMMON MISTAKES

- × Forgetting to use `await` with `all()` or `allInnerTexts()`
- × Not specifying `[href]` for links (includes empty anchors)
- × Not specifying `[@alt]` for images (includes decorative images)
- × Using wrong loop syntax (use `for...of`, not `for...in`)
- × Forgetting `await` inside loops when getting attributes

★ PRO TIPS!

- 💡 Use `a[href]` to exclude empty anchor tags
- 💡 Use `img[alt]` to get only images with alt text
- 💡 Combine with `filter()` for more specific searches
- 💡 Store count in variables for assertions in tests
- 💡 Use `all()` when you need element references, `allInnerTexts()` for just text

✨ KEY TAKEAWAYS

- Use `allInnerTexts()` to get array of text from elements

- Use `all()` to get array of Locator objects
- Use `.length` to count total elements
- Use `for...of` loop to iterate through arrays
- Always specify attributes like `[href]` and `[@alt]` for accuracy! 🎯

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